AI BOOTCAMP / GROUP 4

Accident anticipation and Amazon review analysis

Input design and model choices, explained through experiments

QBC13 / PROJECT 2

The input representation shapes the experiment

Dashcam video

What evidence is visible before a collision?

- Choose where each clip ends.
- Preserve frame order and useful visual detail.
- Choose joint video learning or a separate temporal model.

Amazon reviews

What information helps predict a rating from 1 to 5?

- Inspect the text and rating distribution.
- Preserve sentiment cues in long reviews.
- Add user context and combine complementary models.

PROBLEM 1: DASHCAM VIDEO

Predicting risk before the accident

1,500

Labeled training videos

1,344

Competition test videos

750 / 750

Accident / normal videos

Prediction horizon	0.5 seconds	1.0 second	1.5 seconds
Input window	Up to 10 seconds	Up to 10 seconds	Up to 10 seconds

The model sees a clip that ends before the event and outputs accident probability.

Source: video reports and problem1 notebooks

Clip boundaries define the learning target

clip_end = event_time - horizon

Choice	Reason
Split videos 80% / 20% before creating clips	Keep every clip from one video in the same split.
Estimate negative-clip positions from training positives	Use a data-driven timing pattern for normal videos.
MViT: select eligible clips at longer horizons	For h > 0.5 s, require alert-to-event lead > h.
Bi-GRU: retain all clips with time-aware labels	Label 1 when clip_end reaches or passes the alert time.

Resulting clip counts: MViT 2,667 / 657 and Bi-GRU 3,600 / 900 (train / validation).

Source: problem1/ai-g4-q1-mvit.ipynb; ai-g4-q1-up.ipynb

MViT input: a compact video tensor



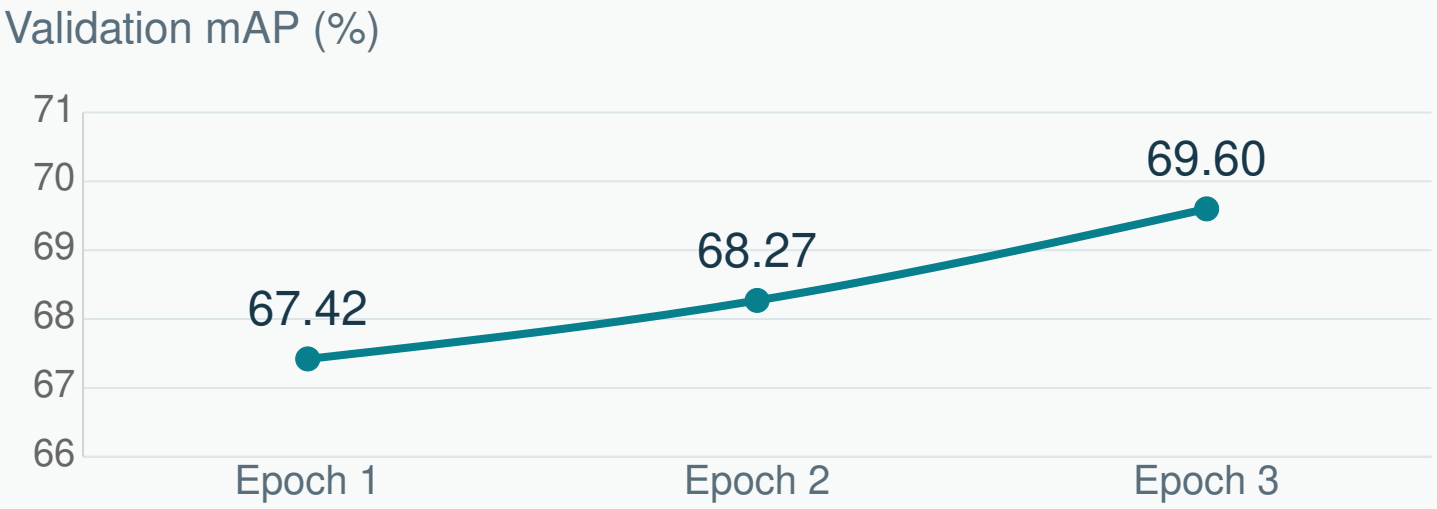
Transformation	Why we chose it
Uniform sampling across the clip	Cover the observation window with a fixed frame budget.
RGB conversion and square resizing	Create the fixed tensor expected by the video backbone.
Scale pixels and use Kinetics-400 normalization	Match the preprocessing associated with pretrained video weights.

The video backbone receives spatial and temporal evidence in one tensor.

Source: MViT V2-S report; problem1/ai-g4-q1-mvit.ipynb

MViT learns spatial and temporal patterns jointly

- Use a pretrained multiscale video transformer to model relationships across frames.
- Preserve pretrained features by freezing early blocks. Train the final block, norm and head.



7.10M

Trainable parameters
out of 34.23M total

Source: MViT V2-S report; recorded training logs

ResNet input: preserve the road view and cache features

Transformation	Why we chose it
24 uniformly sampled frames at 128 x 224	Keep a 16:9 view and an ordered sequence for the temporal model.
Shared brightness and contrast factors per clip	Vary appearance without changing lighting independently frame by frame.
uint8 loading, GPU scaling and ImageNet normalization	Reduce host memory and match the frozen ResNet18 encoder.
Cache one 512-D feature vector per frame	Avoid repeating CNN extraction during temporal-model training.



Source: ResNet18 + Bi-GRU report; problem1/ai-g4-q1-up.ipynb

Bi-GRU models the feature sequence with attention



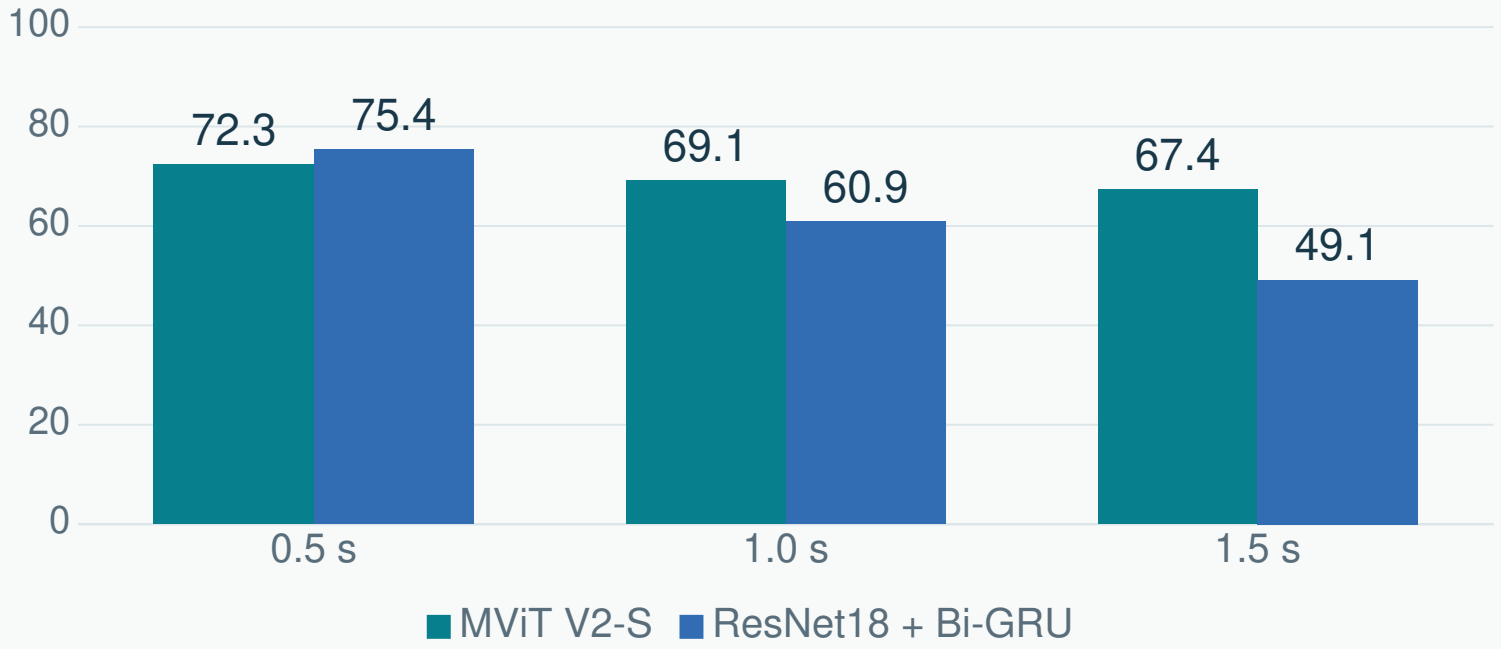
- Bi-GRU combines temporal context within the observed clip. Attention learns how strongly to weight each frame.
- A weighted binary loss accounts for the label mix across prediction horizons.

Pipeline outcome	Recorded value
Trainable temporal model	2,498,050 parameters
Training after feature caching	Less than 1 second per epoch
Selected checkpoint	Epoch 5, mAP = 0.61823

Source: ResNet18 + Bi-GRU report; best checkpoint

The two pipelines produce different horizon profiles

Average Precision (%) on each pipeline's validation clips



0.6960

MViT validation mAP

0.61823

Bi-GRU validation mAP

MViT validation: 657 clips. Bi-GRU validation: 900 clips. Each result follows its own clip-selection and labeling policy.

Source: best validation checkpoints from both video reports

Architecture and input choices side by side

Feature	MViT V2-S	ResNet18 + Bi-GRU
Input	16 frames, 224 x 224	24 frames, 128 x 224
Pretrained knowledge	Kinetics-400 video model	ImageNet frame encoder
Learning strategy	Joint spatial and temporal modeling	Frozen frame features, learned sequence model
Trainable parameters	7.10M	2.50M
Test-video aggregation	Maximum score from 3 clips	Final 10-second clip

Joint video understanding and fast temporal experiments guided the two model choices.

Source: video reports and inference sections

Review preparation made the analysis meaningful

838,944

Original labeled reviews

8,554

Duplicate rows removed

830,390

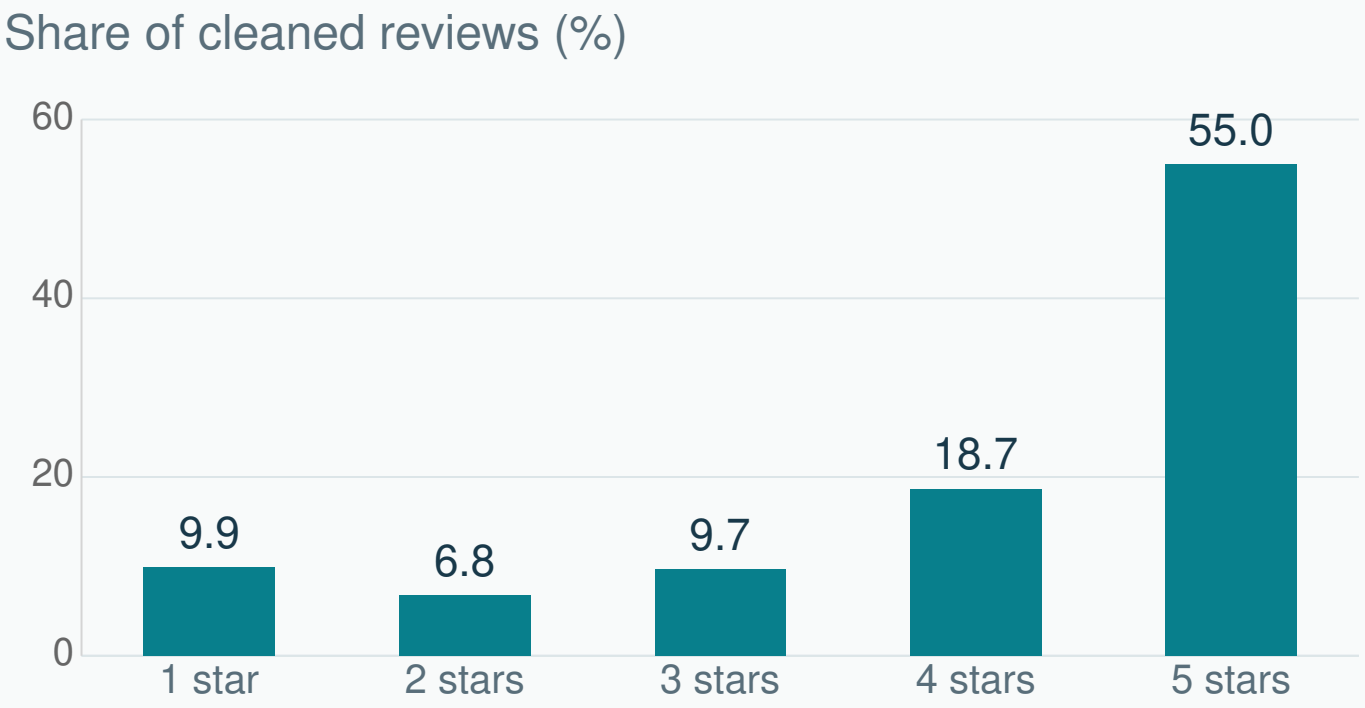
Reviews after cleaning

Preparation	Reason
Deduplicate by user, product and review text	Count repeated observations once.
Join product titles and standardized brands	Interpret product and brand rankings.
Decode HTML and parse numeric vote counts	Create readable text and usable helpfulness totals.

Source: q2_part1_part2.ipynb; Project 2-1&2 report

EXPLORATORY EVIDENCE

Rating imbalance and text length shaped later decisions



409

Median review length (characters)

~1,700

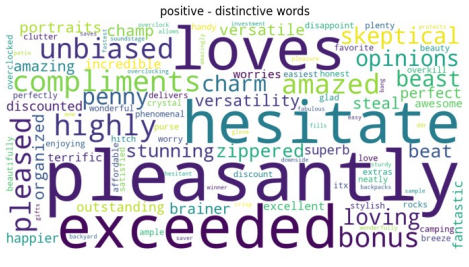
95th-percentile length (characters)

Text length guided the input budget. Rating imbalance guided the final decision rule.

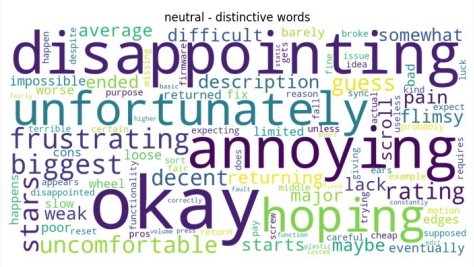
Source: q2_part1_part2.ipynb, rating and length outputs

Distinctive words suggested a warranty question

Positive



Neutral



Negative



4.2x

Relative frequency of "warranty" in negative versus other reviews

This observation motivated aspect analysis: how do warranty-related ratings vary by product?

Source: q2_part1_part2.ipynb, word distinctiveness analysis

User, product and brand views answered different questions

Selected high-volume brand	Mean rating
Anker	4.244
AmazonBasics	4.200
Logitech	3.951
NETGEAR	3.811

Benson Leung

4,572 helpful votes
Top reviewer by total helpfulness

OontZ Angle 3

1,163 five-star reviews
Top product by five-star count

We grouped users by reviewerID and examined brand ratings alongside review volume.

Source: q2_part1_part2.ipynb, ranking tables

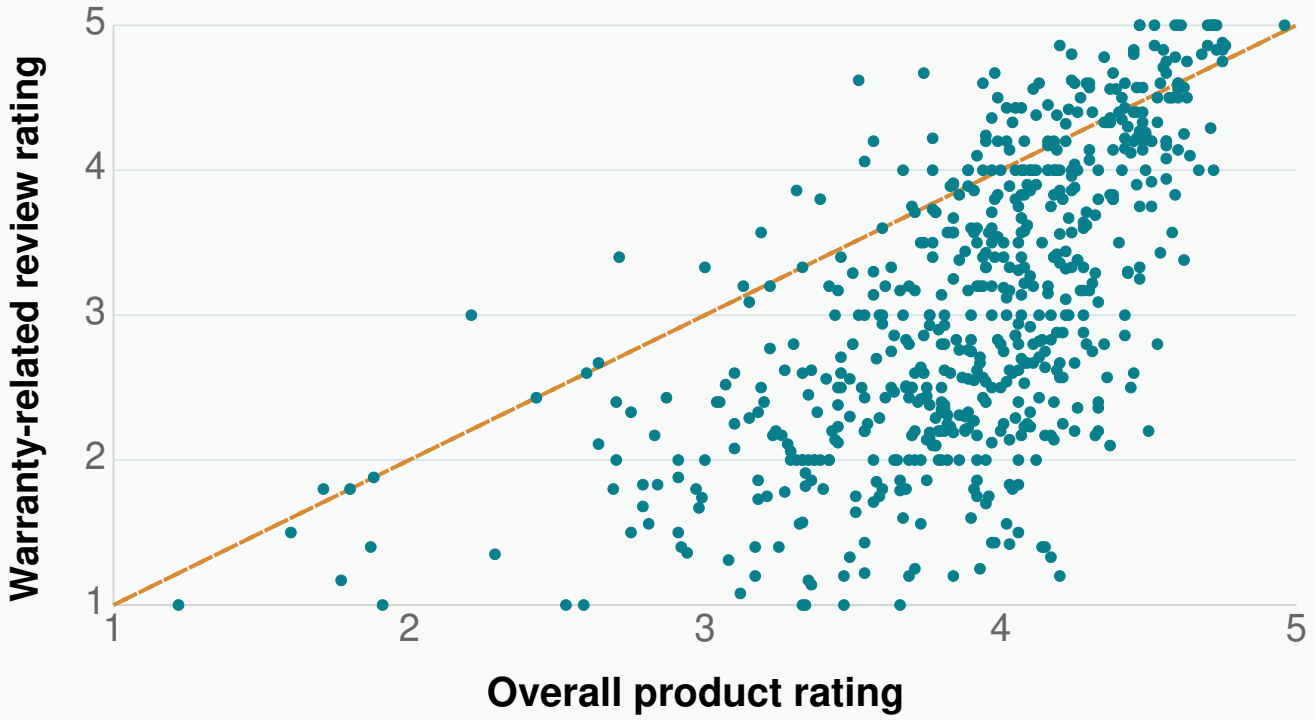
Warranty retrieval used vocabulary from the review domain

Step	Input transformation	Reason
1	Start with warranty / guarantee	Build an explicit keyword baseline.
2	Train Word2Vec and compare with GloVe	Find domain neighbors and observed spelling variants.
3	Expand variants with difflib	Match additional misspellings.
4	Filter non-warranty uses of guarantee	Keep the intended meaning of the word.
5	Include rma after manual inspection	Capture a product-return term used in this domain.

Examples: warrenty, waranty, squaretrade

Source: q2_part1_part2.ipynb, warranty analysis

Warranty-related ratings differ from overall product ratings



18,637

Warranty-related reviews

Initial filter precision: 98% in manual review

Measure	Value
Warranty-review mean	3.08
All-review mean	4.02
Products with 5+ reviews	625

82% of these products lie below the equality line. Their mean product-level rating gap is 0.83 stars.

Source: outputs/warranty_by_product.csv; q2_part1_part2.ipynb

MiniBERT made repeated input experiments practical

11.17M

Approximate parameters

4 layers

4 attention heads per layer

70.54%

V1 validation Micro-F1

- A compact pretrained model allowed repeated full fine-tuning on a T4.
- V1 uses reviewText only, up to 128 tokens, with a new five-class head.
- Dynamic batch padding avoids padding every review to the global maximum.

Controlled experiment sample: 419,472 reviews, split 70% / 15% / 15% for train / validation / internal test.

Source: Project Report.pdf; Mini-Bert/V1_MiniBERT_Baseline.ipynb

V2 preserved both ends of long reviews

Review length	Input policy	Reason
Fits in 256 tokens	Keep the complete reviewText	Use the full review directly.
Exceeds 256 tokens	Add summary, then keep first 192 + last 64 tokens	Retain the introduction and concluding sentiment.
Every batch	Pad dynamically, train with batch size 128	Use a larger batch without a fixed global padding cost.

+1.65 pp

Validation Micro-F1 versus V1

0.3701

MAE, from 0.4122 in V1

20.12 min

Recorded training time V1: 33.68 min

Source: Mini-Bert/V2_Extended_Input.ipynb; Project Report.pdf

V3 tested whether lexical filtering could save token budget

Hypothesis: remove selected function words while preserving sentiment cues.

- Use a predefined 15-word candidate set.
- Protect negation, contractions, intensifiers, sensitive phrases and quoted spans.
- Keep the V2 input policy and model so the new intervention is lexical filtering.

Recorded input effect	Result
Actual model-input tokens	4.21% fewer
Reviews requiring truncation	12.24% to 11.70%
Additional training reviews fitting in full	1,577

Source: Mini-Bert/V3_Stopword_Removal.ipynb; Project Report.pdf

V4 used earlier ratings as an auxiliary user signal

50.2% of external-test rows had strictly earlier history in the full labeled data.

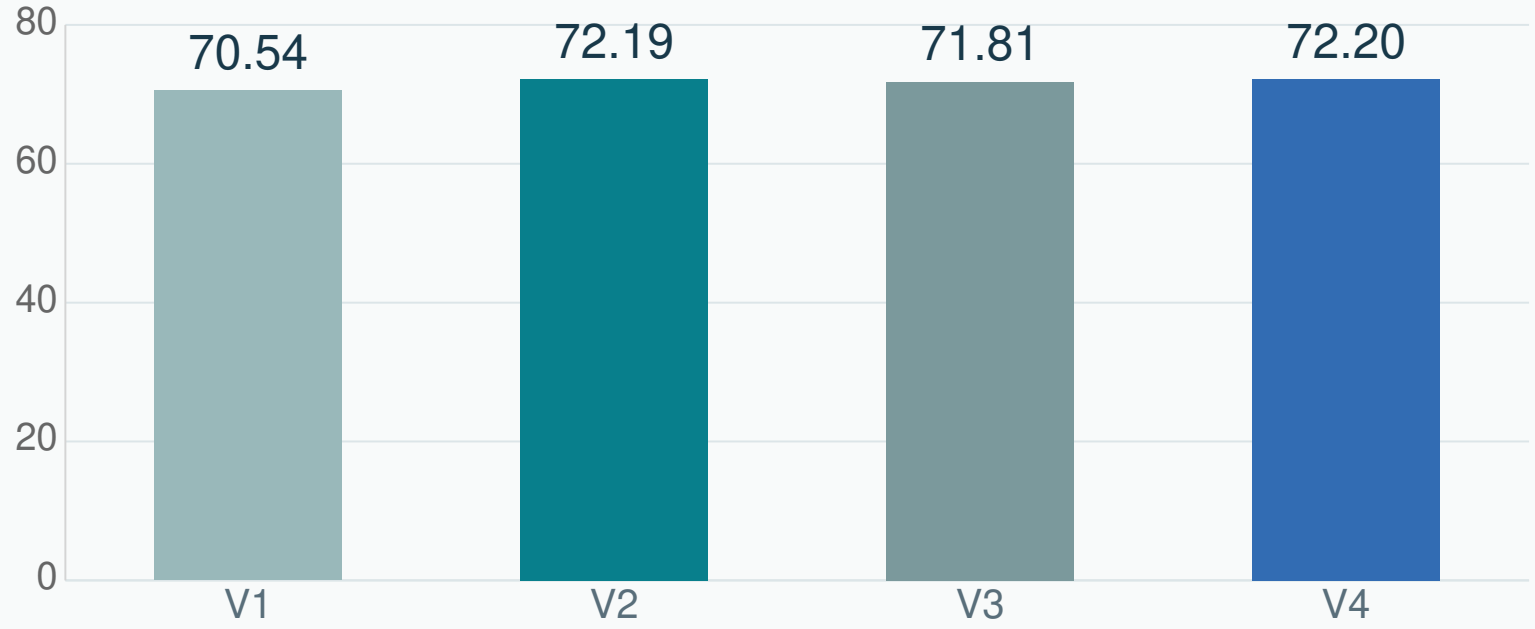
Feature or rule	Why it matters
Mean and count of the user's prior ratings	Encode rating tendency and evidence count.
Only training observations strictly before the target day	Use information available before the target review.
Learn a history distribution and a reliability gate	Learn how much to trust user context.
20% history cap. Text-only fallback.	Keep text as the main evidence source.

p_final = (1 - alpha) p_text + alpha p_history

Source: Mini-Bert/User_History_Feasibility.ipynb; V4_User_History.ipynb

The input experiments produced a measured scorecard

Validation Micro-F1 (%) / common validation split, n = 62,921



+1.65 pp

V2 versus V1

72.20%

V4 validation

V1: text only **V2: richer context** **V3: lexical filtering** **V4: text and user history**

Source: Project Report.pdf; V1-V4 validation outputs

Testing the saved MiniBERT checkpoint

Row-normalized confusion matrix

Actual / Pred	1	2	3	4	5
1	**74.4%**	13.4%	5.9%	1.8%	4.5%
2	34.9%	**28.6%**	25.2%	4.9%	6.4%
3	10.2%	13.1%	**43.7%**	21.5%	11.5%
4	1.6%	1.4%	10.2%	**46.1%**	40.6%
5	0.8%	0.3%	1.3%	6.8%	**90.8%**

72.06%

Internal-test Micro-F1

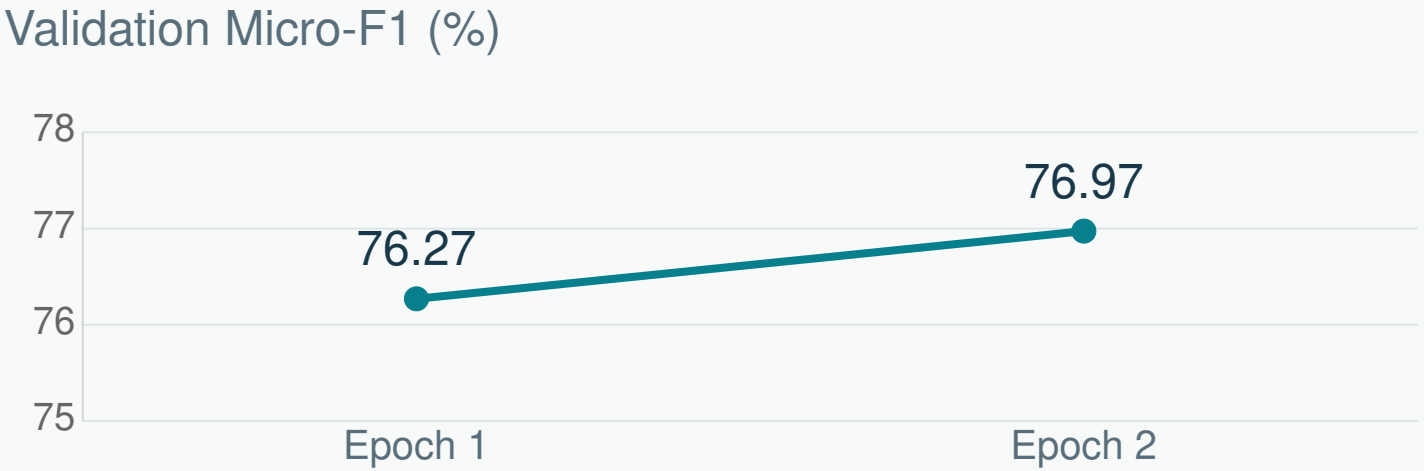
Metric	Value
Test reviews	62,921
MAE	0.3680
5-star recall	90.75%

Rows: actual rating. Columns: predicted rating. All five ratings share the same class labels.

Source: Mini-Bert/V4_Final_Test_Evaluation.ipynb

The final model used a distilled contextual backbone

- Choose a distilled six-layer backbone to keep contextual fine-tuning practical on a T4.
- Carry the text, summary and user-context feature direction into the final modeling stage.
- Fine-tune a five-class classifier and save the selected checkpoint.

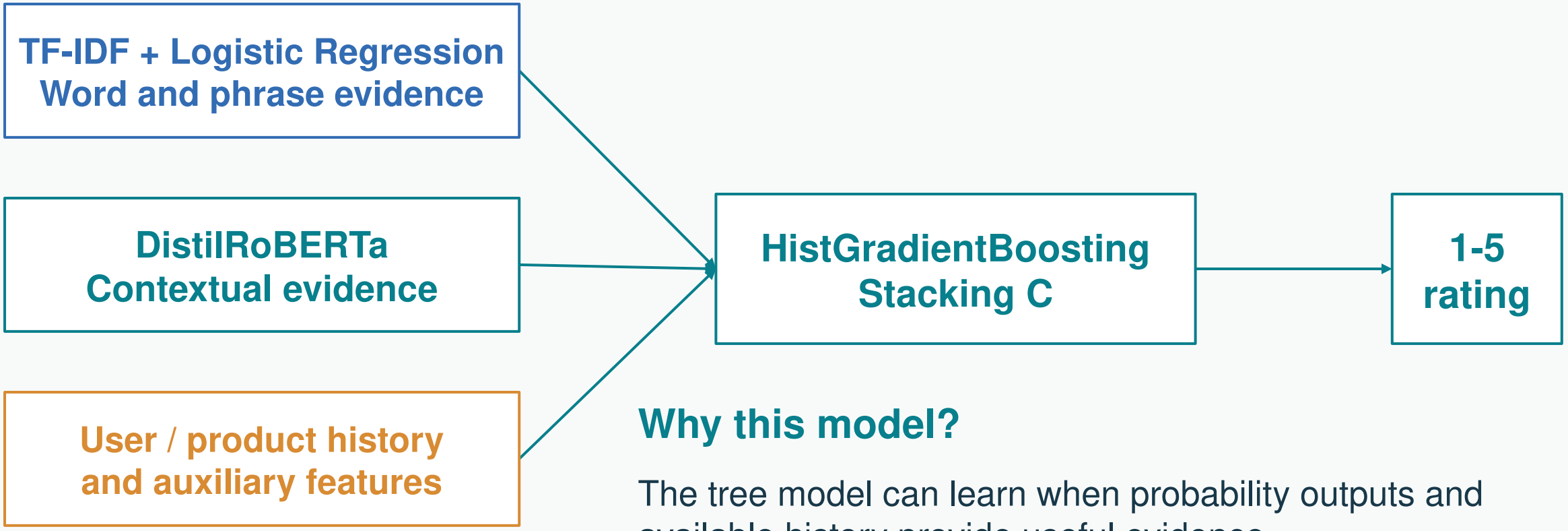


~82M

Parameters in the recorded DistilRoBERTa classifier

Source: problem2/q2_part3_transformer.ipynb, executed checkpoint

Stacking combined complementary evidence

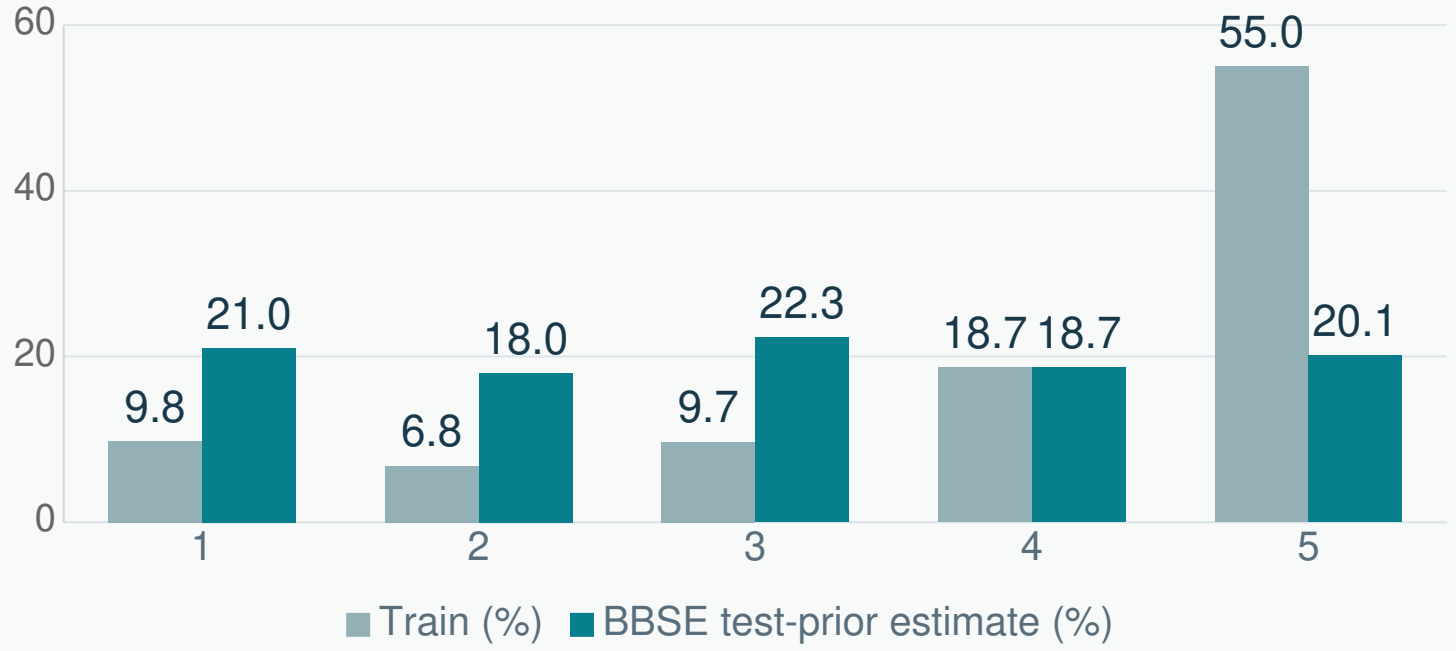


Why this model?

The tree model can learn when probability outputs and available history provide useful evidence.

Source: q2_part3_tfidf_stacking.ipynb, stacking C

Prior correction adapted the final decision rule



p_adjusted(k) proportional to p_model(k) / prior_train(k)^alpha

- BBSE estimated a near-balanced test prior.
- We evaluated the adjusted rule on balanced validation.
- The selected correction uses alpha = 1.

This step adjusts output probabilities after the model. The distribution shown is a BBSE estimate.

Source: q2_part3_tfidf_stacking.ipynb, BBSE and prior correction

Final text-model results by validation setting

Model	Natural validation without correction	Balanced validation with prior correction
TF-IDF + Logistic Regression	70.76%	57.61%
DistilRoBERTa	76.97%	66.26%
Stacking C	77.69%	67.09%

Micro-F1 for the final modeling run in each validation setting

+9.48 pp

Stacking C versus corrected TF-IDF on balanced validation

Source: outputs/part3/results_final.csv

REFERENCE

Project sources and experiment records

Project brief

Team repository

Video notebooks and model reports

Exploration and warranty analysis

MiniBERT experiments and experimental report

Final DistilRoBERTa execution

Stacking and prior correction

Final numeric scorecard

Model identifier in the executed final notebook: distilbert/distilroberta-base

Suggested delivery schedule

Presentation section	Slides	Suggested time	Elapsed window
Opening and project questions	1-2	1 minute	00:00 - 01:00
Video inputs, architectures and results	3-10	10 minutes	01:00 - 11:00
Review exploration and warranty analysis	11-16	6 minutes	11:00 - 17:00
MiniBERT input experiments and results	17-22	7 minutes	17:00 - 24:00
Final backbone, fusion and results	23-26	6 minutes	24:00 - 30:00
Total presentation		30 minutes	

Slide 27 is a reference page. This schedule is for rehearsal and stays outside the main presentation.